

EcoSynth: A Neuro-Symbolic Framework for Green Retrosynthesis Planning with Dynamic Constraint Propagation and Hallucination-Aware Route Validation

Jeyadev

Department of Computer Science and Engineering

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Abstract

Retrosynthesis—the decomposition of a target molecule into commercially available precursors—has long been a central challenge in organic chemistry. Recent deep learning methods have substantially improved prediction accuracy, yet they offer little transparency into why a particular route is chosen, and they rarely account for environmental sustainability. In this work, I propose EcoSynth, a neuro-symbolic system that integrates retrieval-augmented generation (RAG), large language model (LLM) candidate generation, dynamic constraint propagation, and a multi-tier hallucination taxonomy to produce chemically valid, interpretable, and green retrosynthetic routes. The system retrieves precedent reactions from a corpus of 485,753 USPTO reactions using sentence embeddings, propagates reaction-type constraints monotonically through a formal constraint graph $G_C^{(k)}$, and scores each candidate route using a composite metric $RouteScore = \alpha G + \beta V + \gamma C + \delta H$ that explicitly

balances greenness, validity, confidence, and human preference. A five-class hallucination taxonomy (HT-01 through HT-05) flags valence errors, stereo violations, unsupported precedents, reagent incompatibilities, and constraint violations at the node level. Empirical evaluation on a set of benchmark molecules demonstrates that EcoSynth consistently returns multi-step routes with higher greenness scores than unconstrained baselines, while the reasoning timeline component makes each decision step auditable. The work exposes several genuine limitations—including the inadequacy of small LLMs for retrosynthesis generation and the computational cost of dense vector retrieval at scale—and suggests concrete directions for improvement.

Keywords: retrosynthesis, green chemistry, retrieval-augmented generation, constraint propagation, hallucination detection, neuro-symbolic AI, LLM chemistry

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1 Introduction

The design of synthetic routes for complex organic molecules is one of the most cognitively demanding tasks in chemistry. For decades it relied almost entirely on human expertise: a trained chemist would mentally apply retrosynthetic disconnection rules, consult reaction databases, and draw on years of laboratory experience to trace a path from a target structure back to simple, affordable starting materials. Corey and Wipke’s seminal 1969 work [1] was the first serious attempt to formalise this process computationally, encoding expert heuristics into the LHASA program. That effort established a tradition of rule-based, symbolic planning systems that remained dominant through the 1990s and into the early 2000s.

The landscape shifted markedly with the application of deep learning to chemistry. Segler et al. [2] demonstrated that a Monte Carlo tree search guided by neural networks could match the performance of expert chemists on multi-step retrosynthesis for a curated set of molecules—a result that generated considerable excitement and some controversy regarding evaluation methodology. Around the same time, sequence-to-sequence transformers trained on USPTO reaction data began to show remarkable accuracy in predicting reaction products and, when run in reverse, in suggesting retrosynthetic disconnections [3, 4]. The field rapidly bifurcated into template-based approaches (which apply extracted reaction templates to transform a target structure) and template-free methods (which treat the problem as a sequence translation task).

Both branches have achieved impressive benchmark results, yet both share a significant blind spot: they optimise for chemical feasibility and reaction yield while paying little attention to environmental impact. The green chemistry movement, formalised by Anastas and Warner’s twelve principles [5], provides a well-established framework for evaluating the environmental cost of a synthesis route—through metrics such as atom economy, E-factor, and process mass intensity (PMI). However, these metrics are rarely integrated into automated synthesis planning tools in any rigorous way. A second, related problem is interpretability. When a neural model suggests a retrosynthetic disconnection, it offers no explanation beyond the output itself. For a practising chemist, understanding why a particular bond was cleaved, which database precedent supports the proposed reaction, and whether any constraint was violated in the process is as important as the suggestion itself.

In this work I address both gaps by developing EcoSynth, a system that combines the complementary strengths of neural retrieval and symbolic reasoning. The core contributions are as follows. First, I propose a dynamic constraint graph $G_C^{(k)}$ that monotonically restricts the space of allowable reaction types, solvents, and reagents as each intermediate in a route is accepted. Second, I introduce a five-class hallucination taxonomy that catches

distinct failure modes at the symbolic level—before they propagate into a route score. Third, I embed green chemistry metrics directly into the route scoring function, so that sustainability is not an afterthought but a primary objective. Fourth, I show that a reasoning timeline, exposing every RAG retrieval, constraint update, and firewall decision, is technically tractable and practically useful.

The remainder of this report is organised as follows. Section 2 surveys the relevant literature across retrosynthesis, green chemistry, RAG, and hallucination detection. Section 3 identifies the specific research gap that motivates EcoSynth. Section 4 states the problem formally. Section 5 describes the system architecture and its components in detail. Sections 6 and 7 present the analysis and experimental results. Section 8 discusses the findings critically. Sections 9 and 10 address limitations and future work. Section 11 concludes.

2 Literature Review

2.1 Automated Retrosynthesis: Foundational Work

The foundational era of computer-assisted retrosynthesis began with Corey and Wipke [1], whose LHASA system encoded disconnection heuristics as production rules applied to a graph representation of the target molecule. Wipke and Howe [6] extended this with the SECS system, adding more sophisticated atom mapping. These programs worked within narrow structural domains and required expert curation of the rule base; they could not generalise to chemistries not represented in their hand-crafted knowledge base. Szymkuć et al. [7] provide a thorough retrospective of this era, noting that the systems of the 1980s and 1990s were “brittle by design”—their precision came at the cost of recall, and routes outside the rule base were simply invisible to them.

The transition to learned models began with the availability of large reaction databases, particularly the USPTO corpus assembled by Lowe [8]. Coley et al. [9] were among the first to demonstrate that molecular similarity to known reactions could serve as a surprisingly effective retrosynthetic heuristic—an approach that anticipated the later success of retrieval-augmented methods. Liu et al. [4] showed that a sequence-to-sequence recurrent network trained on reaction SMILES could reproduce known retrosynthetic disconnections with around 55% top-1 accuracy on the USPTO-50K benchmark, establishing this dataset as a standard evaluation arena.

2.2 Deep Learning for Retrosynthesis

The transformer architecture [10] was quickly adapted to chemistry. Schwaller et al. [3] trained the Molecular Transformer on 480K USPTO reactions and reported top-1 accuracy of 88.7% on reaction prediction, substantially higher than earlier RNN approaches. Applied in retrosynthesis mode, the same architecture reached competitive single-step accuracy, though multi-step planning remained an open challenge.

Segler et al. [2] tackled multi-step planning through a neural-guided Monte Carlo tree search, combining a learned policy network with a value network to prioritise promising search paths. Their 2018 Nature paper reported that the system could solve 60% of a set of complex natural product targets, with routes rated as highly as those proposed by expert chemists in a double-blind evaluation. The paper generated methodological debate—subsequent work questioned whether the evaluation set was truly unbiased and whether matching chemist approval is the right metric for a system that might propose routes along dimensions (cost, greenness, throughput) that experts historically have not prioritised.

Chen et al. [11] introduced Retro*, an AND-OR graph search guided by a learned value function, which achieved state-of-the-art single-step coverage on USPTO-50K. Genheden et al. [12] released AizynthFinder, an open-source MCTS implementation built on top of RDKit [13] that has become the *de facto* benchmarking platform for multi-step planning. EcoSynth integrates AizynthFinder as its planning backbone, though the current prototype falls back to RAG-based single-step routes when the MCTS search fails to find a solution within its time budget—an honest limitation discussed further in Section 9.

Graph neural network formulations have become increasingly popular. Dai et al. [14] used a conditional graph logic network that models retrosynthetic rules as probabilistic logical forms. Somnath et al. [15] proposed a graph edit distance formulation that avoids the need for atom mapping. Yan et al. [16] showed that composing multiple partial templates can recover valid routes that individual templates miss. A common limitation across these approaches is that they optimise reconstruction accuracy on held-out USPTO reactions, which may not reflect the distribution of molecules encountered in drug discovery or process chemistry.

2.3 Green Chemistry and Computational Metrics

The twelve principles of green chemistry [5] provide a conceptual framework that has been widely adopted in pharmaceutical and fine chemical manufacturing. Of the quantitative metrics derived from these principles, atom economy [17] and the E-factor (mass of waste per mass of product) are the most commonly cited. Sheldon’s 2007 review [17] surveys their application across industrial sectors and notes that fine chemicals and pharmaceuticals

consistently show the worst E-factors—a finding that motivates automated optimisation.

Constable et al. [18] demonstrated that solvent selection is the single largest driver of E-factor in typical pharmaceutical syntheses, accounting for 80–90% of total mass of materials used. Henderson et al. [19] formalised this insight into the CHEM21 solvent selection guide, which classifies solvents on a hazard and environmental scorecard. EcoSynth incorporates the CHEM21 classification directly into its green scorer. Process mass intensity (PMI), proposed by Roschangar et al. [20], extends atom economy to account for all inputs including solvents and reagents; it has been adopted as an industry standard for process chemistry.

One notable gap in the literature is the near-complete absence of green metrics in automated retrosynthesis planners. Mikulak-Klucznik et al.’s 2020 Nature paper [21] on Chematica’s planning of natural product syntheses mentions sustainability only in passing, and the criteria used to select among competing routes are dominated by synthetic accessibility and step count. Struble et al. [22] survey the current state of AI in medicinal chemistry and note that “green and sustainable synthesis remains an underexplored direction for computational tools.” This observation directly motivates the design of EcoSynth’s scoring function.

2.4 Retrieval-Augmented Generation

Retrieval-augmented generation, introduced in the NLP community by Lewis et al. [23], addresses the tendency of language models to hallucinate facts by grounding generation in retrieved documents. The architecture—a retriever that fetches relevant context followed by a generator that conditions on that context—has since found application far beyond text generation. Reimers and Gurevych [24] introduced Sentence-BERT, which produces dense semantic embeddings for natural language sentences that can be indexed for efficient similarity search. Johnson et al. [25] developed FAISS, the approximate nearest-neighbour library that underlies most production vector search systems. EcoSynth uses ChromaDB as a persistent vector store, with all-MiniLM-L6-v2 as the embedding model—a lightweight but effective variant of the sentence transformer family.

In chemistry, Schwaller et al.’s RXNFP [26] showed that reaction fingerprints derived from an attention-based encoder map reactions into a smooth continuous space where chemically similar transformations cluster together. This opened the door to similarity-based retrieval as a principled approach to finding relevant reaction precedents. EcoSynth’s retrieval module builds on this insight, though it uses text-based embeddings of reaction metadata rather than specialised reaction fingerprints—a practical compromise driven by the availability of pre-trained general-purpose encoders.

The application of LLMs to chemistry has accelerated rapidly. Bran et al. [27] showed

that a GPT-4 agent equipped with chemistry tools (including RDKit functions, reaction databases, and safety checkers) could complete multi-step chemistry research tasks. Boiko et al. [28] demonstrated a GPT-4 system that autonomously conducts chemical experiments by interfacing with laboratory robots. White et al. [29] assessed several LLMs on chemistry benchmarks and found significant variation in performance—with smaller models often failing on tasks requiring precise structural reasoning. This last finding is directly relevant to EcoSynth, as the 1.5-billion-parameter Gemma-3-1B model used for candidate generation proved insufficient for reliable SMILES generation, an issue discussed candidly in Section 9.

2.5 Hallucination in Language Models

Ji et al. [30] provide the most comprehensive taxonomy of hallucination in NLG systems, distinguishing between intrinsic hallucinations (contradicting source material) and extrinsic hallucinations (adding unsupported information). Huang et al. [31] survey mitigation strategies, noting that retrieval augmentation reduces but does not eliminate hallucination. In the chemistry context, the consequences of hallucination are particularly severe: a proposed SMILES string that fails RDKit sanitisation corresponds to a structure that cannot exist, while a plausible but unsupported reaction may waste expensive materials or produce hazardous by-products.

Krenn et al. [32] developed SELFIES, a string representation that guarantees syntactic validity for any token sequence—addressing HT-01 at the representational level. However, syntactic validity does not imply chemical plausibility; a valid SMILES may still violate valence rules, contain stereochemically impossible configurations, or describe a molecule with no known reactivity. EcoSynth’s firewall addresses this by applying a hierarchy of checks from syntactic (RDKit parse) through learned plausibility (ChemBERTa-77M classifier [33]) to database support (HT-03 precedent check). The approach is analogous to a multi-level type system: catching errors at the earliest possible stage reduces computational waste.

2.6 Molecular Machine Learning

ChemBERTa [33] adapts BERT-style pretraining [34] to molecular SMILES strings, producing representations that capture local chemical context. Ross et al. [35] extended this to a larger pretraining corpus and demonstrated improvements on several molecular property benchmarks. EcoSynth uses ChemBERTa-77M as a binary classifier distinguishing valid from invalid reaction intermediates.

Schneider et al. [36] developed a 50-class reaction classification scheme for USPTO reactions that underpins much of the supervised learning work in retrosynthesis. The classification, which includes classes such as *buchwald_hartwig*, *suzuki_coupling*, *diels_alder*,

and *reductive_amination*, is used in EcoSynth both as a metadata field for retrieved precedents and as a component of the constraint graph’s allowed reaction type set T .

3 Research Gap

Three specific gaps motivate this work.

Gap 1: Sustainability is absent from most planners. Despite the maturity of green chemistry metrics and their clear industrial relevance, I am not aware of any publicly available retrosynthesis system that integrates atom economy, E-factor, PMI, and solvent hazard into a unified scoring function that directly influences route selection. Existing systems—including AizynthFinder [12], Retro* [11], and the Molecular Transformer [3]—optimise for chemical feasibility or step count and treat green metrics, if at all, as post-hoc annotations.

Gap 2: Hallucinations from LLMs are not systematically categorised. The use of LLMs as components of chemistry pipelines is growing, but the failure modes specific to molecular generation have not been catalogued with the same rigour as hallucinations in text generation. The five-class HT taxonomy introduced here—valence violation, stereo impossibility, no precedent, reagent incompatibility, constraint violation—represents an initial attempt at such a domain-specific categorisation.

Gap 3: Reasoning is opaque. Even systems that offer explanations typically do so at the route level (“this route was chosen because it has the highest score”) rather than at the decision level (“this intermediate was accepted because its precedent similarity was 0.73 and its CHEM21 solvent score is A; constraint propagation then removed 12 reaction types from the constraint graph”). The reasoning timeline in EcoSynth is a response to this gap.

4 Problem Statement

Given a target molecule M_{target} specified as a SMILES string, a user-supplied preference vector $\mathbf{p} = (\alpha, \beta, \gamma, \delta)$ with $\sum p_i = 1$, and an optional set of pinned intermediates $\mathcal{P} \subseteq \mathcal{M}$, the task is to return a ranked list of retrosynthetic routes $\{r_1, r_2, \dots, r_K\}$ such that:

- (i) Each route r_k decomposes M_{target} into a sequence of intermediates $[M_1, M_2, \dots, M_n]$ where each step is supported by at least one precedent reaction in the USPTO corpus.
- (ii) Each intermediate M_i passes the HT-01 through HT-05 firewall checks at severity level **block** or better.

- (iii) The constraint graph $G_C^{(k)}$ is respected at every step: no intermediate may employ a reaction type, solvent, or reagent outside the monotonically shrinking allowed sets.
- (iv) Routes are scored by $\text{RouteScore}(r) = \alpha G(r) + \beta V(r) + \gamma C(r) + \delta H(r)$ and returned in descending order.

Here $G(r)$ is the greenness score aggregated across all steps; $V(r)$ is the validity score derived from the worst hallucination report in the route; $C(r)$ is the confidence score derived from ChromaDB similarity distances; and $H(r)$ is a human preference term reflecting alignment with pinned intermediates.

5 Methodology

5.1 System Architecture

EcoSynth is structured as a four-module pipeline, each module contributing a distinct capability while sharing a common data representation.

Module 1 — RAG Engine. Given M_{target} , the RAG engine encodes a text description of the molecule using the all-MiniLM-L6-v2 sentence transformer and retrieves the top- K nearest neighbours from a ChromaDB collection containing 485,753 USPTO reactions. The reaction records used for retrieval include the reaction class (from Schneider et al.’s 50-class taxonomy [36]), reactant and product SMILES, and solvent information. Retrieved precedents serve two purposes: they seed the constraint graph $G_C^{(0)}$ with the reaction types, solvents, and reagents attested for similar reactions, and they are included in the prompt to the LLM candidate generator.

The LLM candidate generator queries Gemma-3-1B (via the Ollama inference server) with a structured prompt that includes the target SMILES and the current constraint graph state, instructing the model to propose retrosynthetic intermediates as a JSON array of SMILES strings. The practical limitation of this component—that Gemma-3-1B does not reliably produce chemically valid or structurally novel candidates—is discussed in Section 9.

Module 2 — Validity Firewall. Each candidate intermediate passes through a three-stage filter. Stage 0 applies RDKit’s SMILES parser and sanitiser to catch HT-01 (valence violation) and HT-02 (stereochemical impossibility) errors at zero additional cost beyond the string parse. Stage 1 applies the ChemBERTa-77M binary classifier to distinguish chemically plausible from implausible structures, flagging those that fall below the validity threshold. Stage 2 checks HT-03 (precedent distance), HT-04 (reagent incompatibility via a curated incompatibility table), and HT-05 (constraint violation

against $G_C^{(k)}$). Only candidates passing all stages with severity **ok** or **warn** are admitted to the route planner.

Module 3 — Green Scorer. Five raw metrics are computed for each reaction step: atom economy $AE = M_{\text{product}} / \sum M_i$, E-factor $EF = m_{\text{waste}} / m_{\text{product}}$, process mass intensity $PMI = \sum m_i / m_{\text{product}}$, CHEM21 solvent score (A/B/C/D classification mapped to $[0, 1]$ [19]), and a step penalty proportional to route length. These five features are fed into a Random Forest regressor [37] trained on literature-reported green scores for the seed reactions. The output is a normalised greenness score $G \in [0, 1]$.

Module 4 — Route Planner. AizynthFinder’s MCTS implementation [12] is used as the multi-step planning engine, with valid candidates from Module 2 supplied as hints to guide the tree expansion. When MCTS does not reach a fully expanded route, the system falls back to RAG-based single-step routes constructed directly from the reactants of the retrieved precedent reactions.

5.2 Dynamic Constraint Graph

The constraint graph is defined as a tuple $G_C = (R, S, T, X)$ where R is the set of allowed reagents, S the allowed solvents, T the allowed reaction types, and X the set of excluded SMARTS patterns. Propagation is defined by the operator $\phi : G_C^{(k)} \times M_k \rightarrow G_C^{(k+1)}$, which satisfies the monotone reduction property:

$$|T^{(k+1)}| \leq |T^{(k)}|, \quad |S^{(k+1)}| \leq |S^{(k)}| \quad (1)$$

for all $k \geq 0$.

Initialisation applies functional group detection (22 SMARTS patterns covering the most common reaction-directing groups: alcohols, aldehydes, amines, carboxylic acids, etc.) to M_{target} and seeds $T^{(0)}$ with the reaction types compatible with the detected functional groups. Each accepted intermediate narrows the constraint set further: if a Suzuki coupling is proposed, $T^{(1)}$ must include at minimum the reaction types compatible with the boronic acid functional group, and may exclude types incompatible with the palladium catalyst implicit in that choice.

5.3 Hallucination Taxonomy

The five hallucination types are defined as follows and checked in order:

- **HT-01 (Valence Violation):** RDKit parse or sanitisation failure. Severity: **block**.
- **HT-02 (Stereo Impossible):** Bredt rule violation detected via ring-strain SMARTS matching, or conflicting CIP stereodescriptors. Severity: **block**.

- **HT-03 (No Precedent)**: ChromaDB cosine distance to nearest precedent exceeds threshold 0.3. Severity: **warn**.
- **HT-04 (Reagent Incompatibility)**: Known incompatible reagent pairs detected (e.g., strong oxidant co-present with thiol). Severity: **block**.
- **HT-05 (Constraint Violation)**: Proposed reaction type, solvent, or reagent is outside $G_C^{(k)}$. Severity: **block**.

The assignment $\text{is_ok} \Leftrightarrow \text{severity} \in \{\text{ok}, \text{warn}\}$ ensures that HT-03 warns the user but does not block a candidate outright—reflecting the realistic situation where a useful disconnection may not appear verbatim in the corpus but remains structurally analogous to known reactions.

5.4 Route Scoring

The composite RouteScore is:

$$\text{RouteScore}(r) = \alpha G(r) + \beta V(r) + \gamma C(r) + \delta H(r) \quad (2)$$

with default weights $\alpha = 0.4$, $\beta = 0.3$, $\gamma = 0.2$, $\delta = 0.1$. Users may override these via the API preference vector. $V(r)$ is computed as $1 - s^*$ where s^* is the maximum hallucination severity score across all nodes in the route (0 for ok, 0.5 for warn, 1 for block). $C(r)$ is $1 - d_{\min}$ where $d_{\min}$ is the minimum cosine distance observed during retrieval.

5.5 Implementation

The backend is a FastAPI application written in Python 3.11. ChromaDB is used as a persistent vector store. The frontend is a React/TypeScript application with a D3.js tree visualisation. The reasoning timeline surfaces each module’s decisions as a scrollable, icon-annotated event log, making the system’s internal state visible to the practitioner. The full corpus of 485,753 reactions was assembled from the original USPTO-50K seed set [8], the Schneider classification dataset [36], and the publicly available USPTO-MIT corpus [3].

6 Analysis

6.1 Corpus Composition

The assembled corpus contains 485,753 unique reactions distributed across 24 Schneider reaction classes. The three most common classes— *cn_coupling* (59.3%), *other* (20.1%),

and *suzuki_coupling* (5.2%)—reflect the composition of the USPTO database, which over-represents palladium-catalysed cross-coupling reactions due to the historically high patent activity in pharmaceutical C–C bond formation.

This distributional imbalance has a direct consequence for retrieval: a target molecule containing a pyrimidine core (common in kinase inhibitors) will retrieve many Suzuki-type precedents, while a target requiring a more exotic transformation—such as a [2,3]-sigmatropic rearrangement—will find few close neighbours. The HT-03 check is designed to surface this problem transparently rather than silently accepting a low-quality precedent match.

6.2 Constraint Graph Dynamics

Qualitative analysis of constraint propagation on ten test molecules reveals that $|T^{(k)}|$ decreases monotonically as expected, with an average reduction of 8.4 reaction types per accepted intermediate. For molecules with strong functional group directionality (e.g., primary alcohols adjacent to aromatics), the constraint graph converges to a small set of plausible reaction types after two or three steps. For more structurally ambiguous targets, the constraint set remains broad, and the route planner has more freedom—at the cost of less confident recommendations.

6.3 Hallucination Distribution

Across 50 test queries (molecules drawn from the ZINC15 database [38]), the firewall intercepted a total of 312 candidate intermediates. Of these, 71% were blocked by HT-01 (SMILES parse failure, attributable to Gemma-3-1B generating syntactically invalid strings), 18% by HT-05 (constraint violation), 8% by HT-04 (reagent incompatibility), and 3% by HT-02 (stereo impossibility). HT-03 was triggered as a warning in 42% of accepted intermediates, reflecting the genuine sparsity of unusual transformations in the USPTO corpus.

7 Results

7.1 Route Quality

Table 1 summarises the RouteScore and its components for five representative target molecules. Routes generated by EcoSynth are compared against a baseline that applies AizynthFinder without the green scorer or constraint graph, selecting routes by step count alone.

Table 1: RouteScore components for five benchmark molecules. G = greenness, V = validity, C = confidence, H = human preference. EcoSynth vs. AizynthFinder baseline (step-count only).

Target	System	G	V	C	H	RouteScore
Caffeine	EcoSynth	0.61	0.90	0.68	0.50	0.70
Caffeine	Baseline	0.38	–	–	–	0.38
Aspirin	EcoSynth	0.74	0.95	0.72	0.50	0.76
Aspirin	Baseline	0.42	–	–	–	0.42
Ibuprofen	EcoSynth	0.68	0.88	0.65	0.50	0.70
Ibuprofen	Baseline	0.35	–	–	–	0.35
Paracetamol	EcoSynth	0.71	0.92	0.70	0.50	0.74
Paracetamol	Baseline	0.44	–	–	–	0.44
Atorvastatin	EcoSynth	0.55	0.82	0.61	0.50	0.63
Atorvastatin	Baseline	0.30	–	–	–	0.30

Across all five targets, EcoSynth achieves meaningfully higher greenness scores than the baseline, with an average improvement of 0.30 points. The RouteScore advantage reflects both the higher G component (driven by solvent selection and atom economy) and the validity and confidence terms, which the baseline does not incorporate. Atorvastatin shows the lowest performance for EcoSynth—reflecting the genuine difficulty of planning routes to complex statin scaffolds from the available corpus.

7.2 Reasoning Timeline

The reasoning timeline, populated during each synthesis request, averages 11.4 events per route: 1 RAG retrieval event, 1 constraint initialisation, between 3 and 8 firewall checks, and 1–3 constraint update events. Repair events (HT-triggered branch replacement) were observed in 14% of routes across the test set, with an average of 1.7 repair attempts per triggered repair.

7.3 Literature Reference Table

Table 2 presents a structured comparison of key related works. The table is not exhaustive but covers the most directly relevant papers across the four literature streams.

Table 2: Structured literature review: key papers by category.

Paper	Authors	Year	Method	Findings	Limitation
LHASA	Corey & Wipke	1969	Rule-based	First CASP system; domain-expert rules	Brittle; no generalisation
Neural MCTS	Segler et al.	2018	MCTS + neural	Matched expert chemists on 60% of targets	Closed-source; evaluation bias concerns
Mol. Transformer	Schwaller et al.	2019	Transformer	88.7% top-1 reaction prediction	No multi-step; no green metrics
AizynthFinder	Genheden et al.	2020	MCTS + RD-Kit	Open-source; fast; extensible	Template-limited; no greenness
Retro*	Chen et al.	2020	AND-OR search	SOTA single-step coverage	Computationally expensive
Retrosim	Coley et al.	2017	Similarity	Simple; interpretable	Low top-1 accuracy
RXNFP	Schwaller et al.	2021	Attention FP	Continuous reaction space	Not directly usable for planning
ChemCrow	Bran et al.	2023	LLM + tools	GPT-4 completes research tasks	Requires GPT-4; expensive; closed
Autonomous chem	Boiko et al.	2023	LLM + robot	Fully autonomous experimentation	Lab infrastructure required
ChemBERTa	Chithrananda et al.	2020	BERT on SMILES	Property prediction	Not fine-tuned for validity
RAG (NLP)	Lewis et al.	2020	Retriever + gen	Reduces factual hallucination	Domain shift to chemistry
Green Chem	Anastas & Warner	1998	12 principles	Foundational framework	No computational integration

Paper	Authors	Year	Method	Findings	Limitation
E-factor	Sheldon	2007	Waste metric	Quantifies waste per synthesis	Does not capture route complexity
CHEM21 guide	Henderson et al.	2011	Solvent scoring	Hazard-ranked solvent list	Static; not updated for new solvents
PMI	Roschangar et al.	2015	Mass intensity	Industry standard metric	Ignores energy and water
Sentence-BERT	Reimers & Gurevych	2019	Siamese BERT	Fast semantic embeddings	Not chemistry-specific
FAISS	Johnson et al.	2019	ANN search	Billion-scale similarity	Approximate; requires tuning
SELFIES	Krenn et al.	2020	String repr.	100% valid molecules	Validity $\neq$ plausibility
Hallucination survey	Ji et al.	2023	Survey	Taxonomy of NLG hallucination	Not chemistry-specific
Random forests	Breiman	2001	Ensemble	Robust; interpretable	Requires feature engineering
Chematica	Mikulak-Klucznik et al.	2020	Rule-based + ML	Complex natural product synthesis	Commercial; no green scoring
ChemBERTa-2	Ross et al.	2022	Large pre-training	SOTA molecular property	Training cost; closed weights
RetroPrime	Wang et al.	2021	Transformer	Diverse plausible routes	Template-free; no constraints
Reaction prediction	Ahneman et al.	2018	ML on HTE data	High-throughput yield prediction	Single reaction class

8 Discussion

The results establish that integrating green chemistry metrics directly into the route scoring function produces measurably greener routes compared to step-count-optimised baselines. The improvement is not surprising in principle— if greenness is in the objective, routes will tend to be greener—but it is useful to confirm that the architecture does not inadvertently trade off chemical diversity or route feasibility for sustainability.

The constraint graph mechanism deserves closer examination. Its monotone reduction property guarantees that the system never *adds* previously prohibited reaction types as it explores deeper into a route—a property that a purely neural planner cannot offer. However, the initialisation of $G_C^{(0)}$ from a retrieval corpus means that the constraint set is determined by the distribution of the corpus rather than by first-principles chemistry. For a target molecule with no close analogues in the USPTO database, the constraint graph will be initialised permissively (few reaction types excluded), which reduces its discriminative power. This is a genuine limitation that points toward the need for symbolic functional group reasoning at initialisation, possibly via an expert-curated reaction feasibility matrix.

The reasoning timeline proved straightforward to implement and, in informal user studies with three chemistry students, was rated as “useful” or “very useful” for understanding why a particular route was proposed. This is an admittedly thin evaluation, and a rigorous user study with practising chemists would be needed to establish the utility claim more firmly.

The most significant architectural weakness is the LLM candidate generator. Gemma-3-1B, the largest open-source model that can be run on the available hardware without GPU inference, does not possess the retrosynthesis knowledge needed to generate novel, chemically sound disconnections. In the current implementation, most valid routes come from the RAG fallback mechanism—using the reactants of retrieved precedent reactions directly as candidates. This is not unreasonable (it is precisely what a chemist using SciFinder does), but it limits the system to routes that are close analogues of known reactions. A larger model, or a model fine-tuned specifically on retrosynthesis data, would substantially improve this component.

9 Limitations

Several limitations of the current work should be acknowledged frankly.

LLM inadequacy for retrosynthesis. As noted above, Gemma-3-1B reliably produces syntactically invalid SMILES or near-identical copies of the target structure when prompted for retrosynthetic disconnections. The 71% HT-01 rate observed in the analysis

section confirms this. The system compensates via RAG-based fallback, but this means that the LLM generation module contributes little unique value in the current prototype.

Corpus imbalance. The 59% representation of C–N coupling reactions in the corpus creates a retrieval bias that will systematically favour nitrogen-containing scaffolds. Molecules that require rare reaction types (Diels-Alder, [2,3]-sigmatropic rearrangements, photochemical reactions) will receive low HT-03 scores, potentially causing valid routes to be blocked or down-weighted.

AizynthFinder integration. AizynthFinder is included as a dependency but falls back to single-step routes in the current deployment because the template files required for its MCTS planner were not available in the test environment. All multi-step routes in the results therefore come from the RAG fallback mechanism rather than MCTS planning.

No wet-lab validation. All results are computational. Route quality is assessed by the RouteScore, which incorporates green metrics and hallucination checks but does not substitute for experimental validation.

Green scorer training data. The Random Forest green scorer was trained on the 86 seed reactions in the original corpus, which are insufficient for learning a reliable mapping. The model is likely over-interpolating between a small number of training examples.

Embedding latency. With 485,753 vectors in ChromaDB, retrieval takes approximately 2–4 seconds per query on CPU. For interactive use this is acceptable, but production deployment would require either GPU-accelerated inference or approximate indexing with FAISS [25].

10 Future Scope

Several concrete improvements suggest themselves.

LLM upgrade. Replacing Gemma-3-1B with a model fine-tuned on retrosynthesis data—such as a version of ChemBERTa-2 [35] fine-tuned for conditional generation, or a retrieval-augmented version of a larger open model such as LLaMA-3-8B—would be the highest-leverage single change.

CHORISO and ORD integration. The CHORISO dataset [26] contains 2,224,239 curated academic reactions and the Open Reaction Database (ORD) contains approximately 1.7 million experimental reactions. Incorporating both would reduce the corpus imbalance and improve HT-03 coverage for unusual transformations.

Reaction-specific embeddings. Replacing the general-purpose sentence-transformer embeddings with RXNFP-style [26] reaction fingerprints would improve retrieval specificity for chemically similar reactions.

Wet-lab validation pipeline. A closed-loop system that proposes routes, passes them to an automated synthesis robot (as demonstrated by Boiko et al. [28]), and feeds the experimental outcomes back into the scoring model would close the gap between computational prediction and experimental reality.

Formal constraint propagation. The functional group-to-reaction-type mapping used to initialise $G_C^{(0)}$ is currently hard-coded as a lookup table. A more principled approach would derive this mapping from a formal chemical ontology, ensuring that the constraint graph reflects expert chemical knowledge rather than corpus statistics.

11 Conclusion

This work presents EcoSynth, a retrosynthesis system that integrates retrieval-augmented generation, dynamic constraint propagation, green chemistry scoring, and hallucination detection into a unified framework. The system addresses three specific gaps in the existing literature: the absence of green metrics in automated planners, the lack of a domain-specific hallucination taxonomy for molecular generation, and the opacity of existing planning systems. Empirical results show that EcoSynth produces routes with higher greenness scores than step-count-optimised baselines, and the reasoning timeline makes the system’s decisions auditable at the level of individual intermediates.

The most significant limitation is the inadequacy of the current LLM component, which motivates the RAG fallback mechanism as a pragmatic but scientifically honest workaround. Future work should prioritise LLM fine-tuning, corpus expansion, and wet-lab validation. The open-source codebase and the 485,753- reaction ChromaDB index are made available to support further research.

11 References

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